AMAZON PRIME | PROJECT REPORT DOCUMENT

Amazon Prime Cart Abandonment & Customer Segmentation Analysis

Internship Project Summary Report

Project Overview

This project focuses on analysing customer shopping behaviour on amazon to understand purchase patterns, cart abandonment, customer segmentation, and potential revenue loss. Excel and PostgreSQL were used for data cleaning, analysis, and visualization.

Objectives

- Analyse customer shopping and purchasing behaviour.
- Identify cart abandonment patterns.
- Calculate important business KPIs.
- Compare customer segments and identify improvement opportunities.
- Present findings through an interactive Excel dashboard.

Dataset and Data Preparation

The original amazon dataset contained 1,000 records. After removing 15 duplicate records, 985 records remained. Missing categorical values were handled using "Unknown", and inconsistent text formats were standardized. Potential outliers and unusual values were identified and flagged for further investigation.

Tools Used

- **Microsoft Excel:** Data cleaning, KPI calculations, customer segmentation, and dashboard creation.
- **PostgreSQL:** SQL queries, data analysis, and KPI validation.

Key Results

- Total Sessions: After duplicate removed 985(Original 1000)
- Completed Purchases: 332
- Conversion Rate: 33.71%
- Cart Abandonment Rate: 26.65%
- Checkout and Payment Abandonment Rate: 32.24%
- Average Abandoned Cart Value: 59.17(After not including outliers)

Key Findings

The analysis highlighted customer drop-off during the shopping journey and differences in shopping behaviour across customer segments. Non-Prime customers showed a higher cart abandonment rate than Prime customers. Customer segmentation helped identify browsers, purchasers, and low-, mid-, and high-value abandoners.

Recommendations

1. Simplify the checkout and payment process.
2. Introduce cart reminder notifications.
3. Offer additional benefits to Non-Prime customers.
4. Provide Cash on Delivery with suitable limits and verification.
5. Focus on high-value abandoned carts to reduce potential revenue loss.

Conclusion

This project provided practical experience in data cleaning, Excel analysis, SQL querying, KPI calculation, and dashboard development. The findings can help businesses understand customer behaviour, reduce cart abandonment, and improve conversion rates.

Project Outcome: A cleaned dataset, Excel analysis dashboard, SQL queries, and a summary of actionable business recommendations.